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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Development of an Electronic Differential

A DTL REPORT

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In partial fulfilment of the requirements for the degree of

Bachelor of Engineering in

ELECTRONICS AND COMMUNICATION ENGINEERING

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(Autonomous institution affiliated to VTU, Belagavi)

1.1.1 Department of Electronics and Communication Engineering



CERTIFICATE

Certified that the Design Thinking Lab project titled **Development of an Electronic Differential** is carried out by **Channaveeragouda(1RV21EC048), Chetan (1RV21EC050), Karthik P Shetty(1RV21EC077), Mallikarjuna A L (1RV21EC094)** who are the bonafide student of RV College of Engineering, Bengaluru, in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi during the year 2020-2021. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the Design Thinking Lab report deposited in the departmental library. The Design Thinking Lab report has been approved as it satisfies the academic requirements in respect of Design Thinking Lab work prescribed by the institution for the said degree.

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DECLARATION

We **Channaveeragouda, Chetan, Karthik P Shetty, Mallikarjuna A L** students of fourth semester B.E., Department of Electronics and Communication Engineering, RV College of Engineering, Bengaluru, hereby declare that the Design Thinking Lab project titled '**Development of an Electronic Differential**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering in Electronics and Communication Engineering** during the year 2020-2021.

Further, we declare that the content of the report has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this Design Thinking Lab carried out at RVCE will be the property of R V College of Engineering, Bengaluru and we will be one of the authors of the same.

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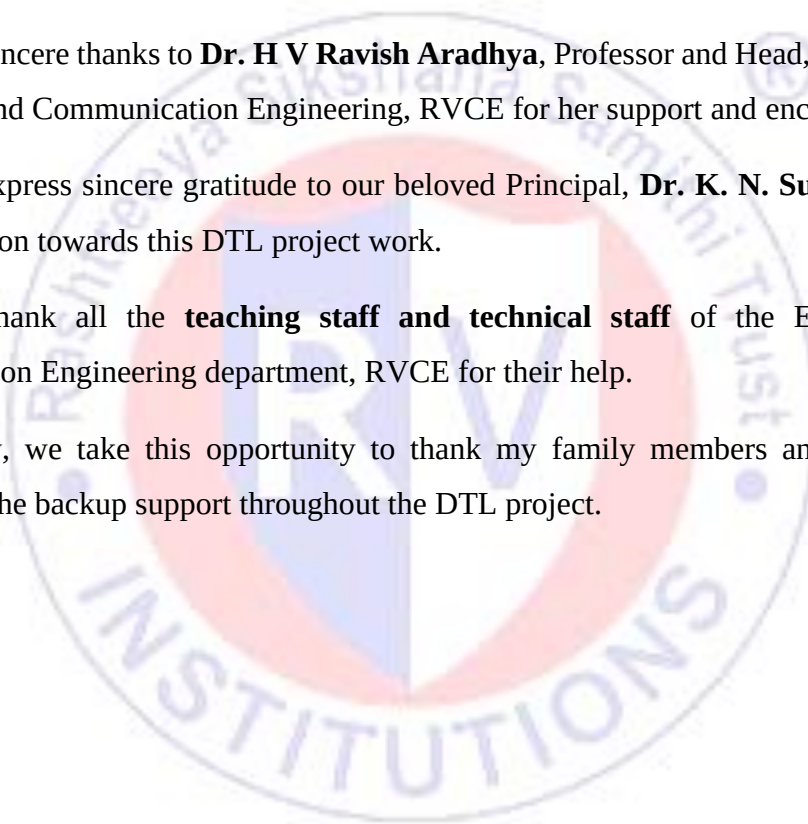
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ABSTARCT

An electronic differential, also known as an e-differential or electronic limited-slip differential (e-LSD), is a technology used in modern vehicles to optimize traction and handling by intelligently distributing torque between the wheels. Unlike traditional mechanical differentials, which rely on purely mechanical components to distribute power, electronic differentials use sensors, computers, and actuators to achieve their functionality.

The primary goal of an electronic differential is to manage the distribution of torque between the left and right wheels of a vehicle, particularly in situations where there is a difference in traction between the two wheels. This occurs during turns, acceleration, or when driving on slippery surfaces. By adjusting torque distribution, the electronic differential can enhance stability, cornering performance, and overall vehicle control. Here's a basic overview of how an electronic differential works:

1. **Sensors:** Various sensors are placed around the vehicle to monitor factors such as wheel speed, vehicle speed, steering angle, acceleration, and more. These sensors provide real-time data to the vehicle's control system.
2. **Control Unit:** The control unit, often a part of the vehicle's electronic stability control (ESC) system, processes the data from the sensors to determine the vehicle's current driving conditions, such as whether it is turning, accelerating, or encountering slippery surfaces.
3. **Torque Distribution:** Based on the information received from the sensors, the control unit calculates the optimal torque distribution between the wheels. If one wheel is slipping or losing traction, the control unit can direct more torque to the wheel with better traction.
4. **Actuators:** To adjust torque distribution, the control unit uses actuators, which can be electromechanical clutches or braking systems. These actuators can apply more force to one wheel while limiting the torque to the other, effectively controlling the power distribution and mitigating wheel spin.
5. **Benefits:** The electronic differential provides several benefits, including improved traction, enhanced stability, better cornering performance, reduced understeer and oversteer tendencies, and increased overall control in various driving conditions.

6. Modes: Some vehicles equipped with electronic differentials might offer different driving modes that allow the driver to choose between various settings, such as "normal," "sport," or "off-road." Each mode adjusts the electronic differential's behaviour to suit different driving scenarios.

Overall, electronic differentials contribute to safer and more dynamic driving experiences by optimizing torque distribution and improving the vehicle's handling characteristics. They are commonly found in high-performance vehicles, sports cars, SUVs, and some advanced all-wheel-drive systems.



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CHAPTER 1

EMPATHY

CHAPTER 1

EMPATHY

- Empathizing means putting yourself in the shoes of the user or customer.
- It involves listening and observing to understand their needs, wants, and feelings.
- Empathy helps designers create solutions that meet user needs and expectations.
- It requires seeking out diverse perspectives and experiences.
- Empathize is a mindset that involves being open, curious, and non-judgmental.
- Techniques such as interviews, observations, surveys, and user personas can help designers

practice empathy

: 1.1 Problem Statement:

The conventional mechanical differentials in vehicles often face limitations in distributing torque effectively between the wheels, especially in scenarios with varying traction conditions. This can lead to reduced maneuverability, traction loss, and compromised stability, particularly in off-road, slippery, or high-performance driving situations. To address these challenges, there is a pressing need for an advanced electronic differential system that can intelligently manage torque distribution between wheels in real-time

1.2 Survey analysis:

A survey was conducted among students, professionals like engineers working in companies, and teachers about the ECG topic and several responses were recorded and studied. Some of the questions asked were:

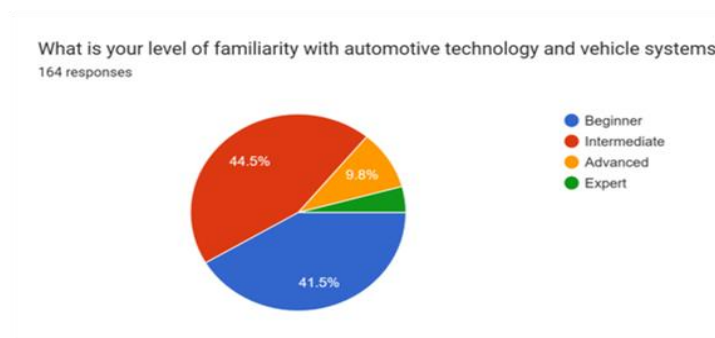


Fig.1.1 Survey graph 1

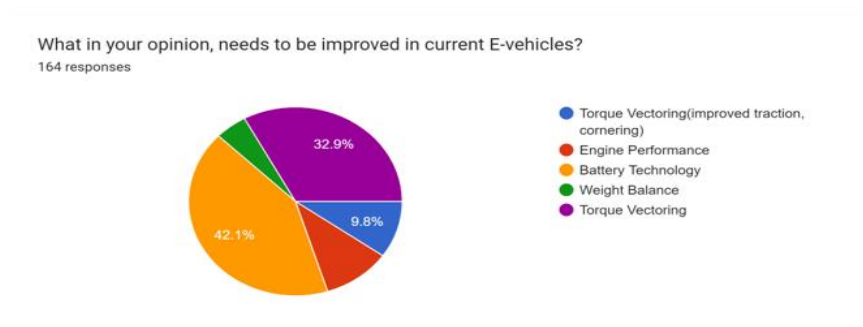


Fig.2 Survey graph 2

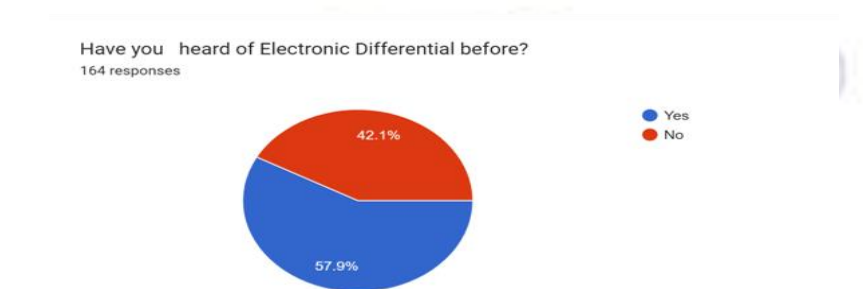


Fig.1.3 Survey graph 3

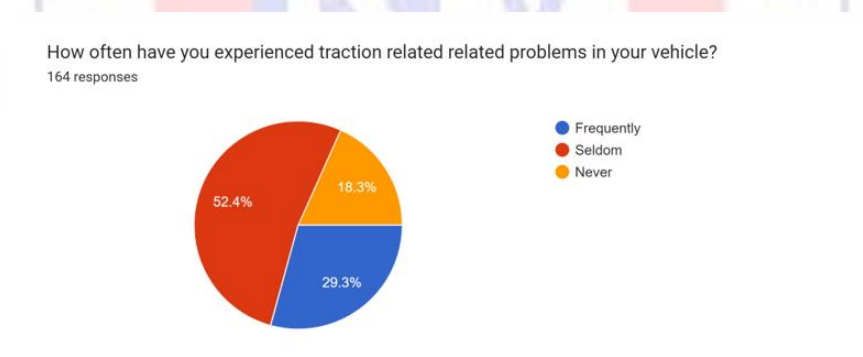


Fig.4 Survey graph 4

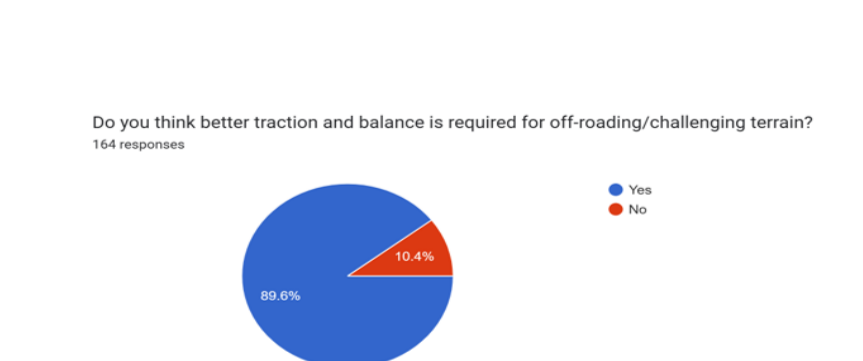


Fig.1.5 Survey graph 5

Do you think E-differential will revolutionize automotive industry?
164 responses

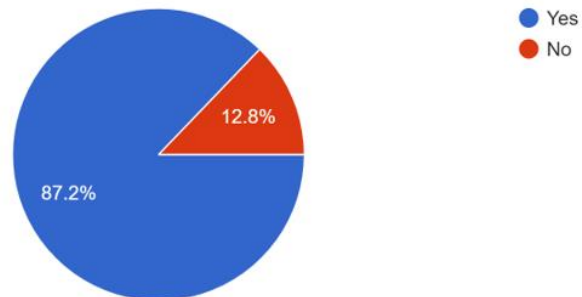


Fig.1.6 Survey graph 6

1.3 Best Responses and suggestions given by responders:

I suggest you go through the papers and models of students from IIT-Bombay/Madras on similar experiments and projects so that you can have a clear view and knowledge about the future of this Exp.

Making use of e-diff for carwalk like hummer or sliding motion which could help in easy turning in ghats and sliding on off road vehicles which could help to avoid obstacles just by sliding left or right instead of taking back and struggling to turn .
This could also reduce the turning radius of even bigger vehicles .proper optimisation of e diff could help in easy overtaking and use e- diff for braking

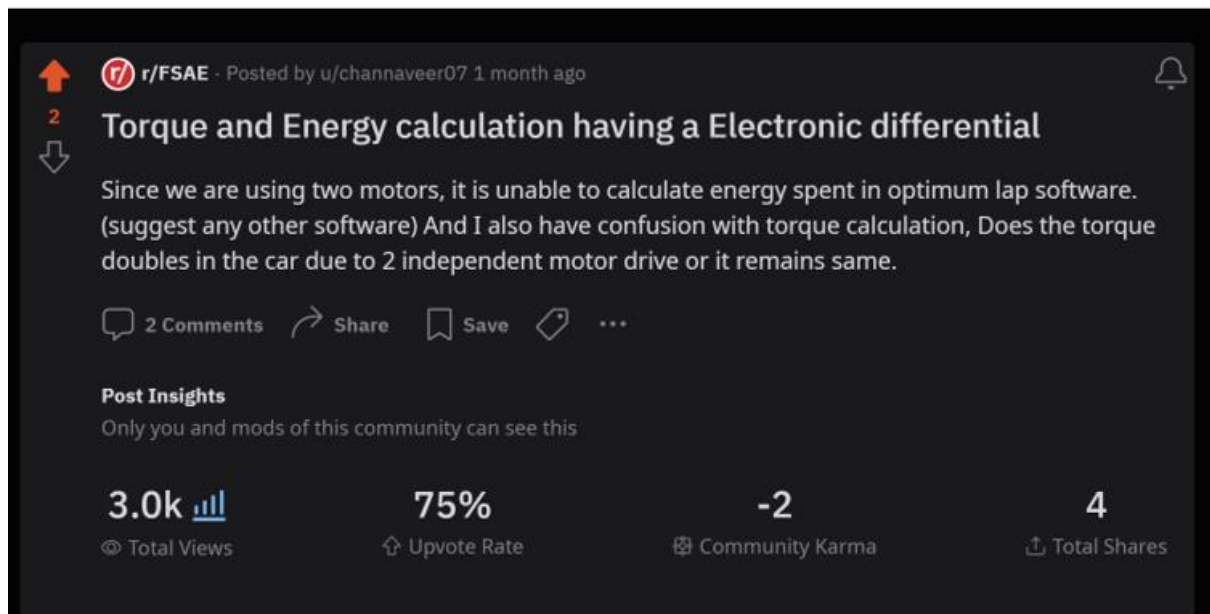


Fig..1.7Survey graph 7

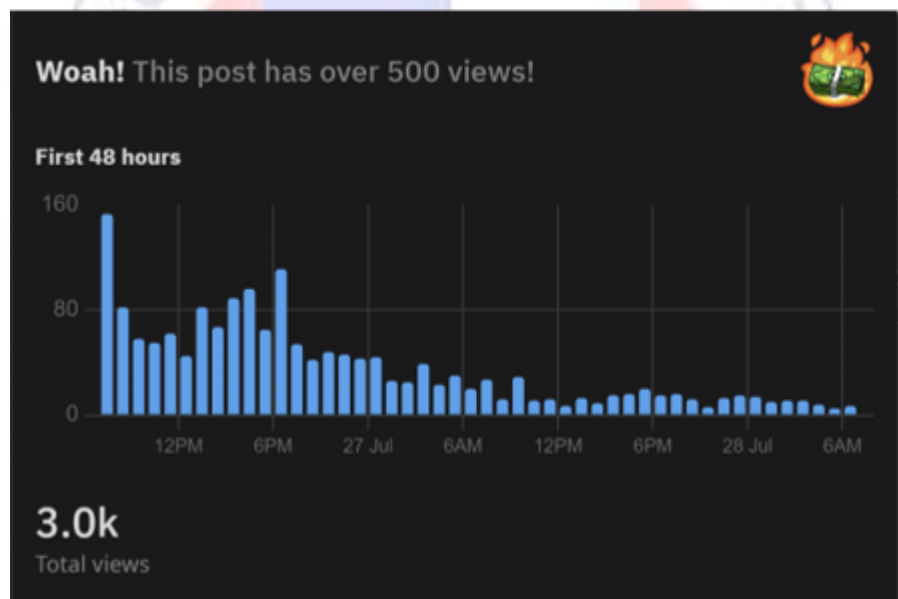


Fig.1.8 Survey graph 8

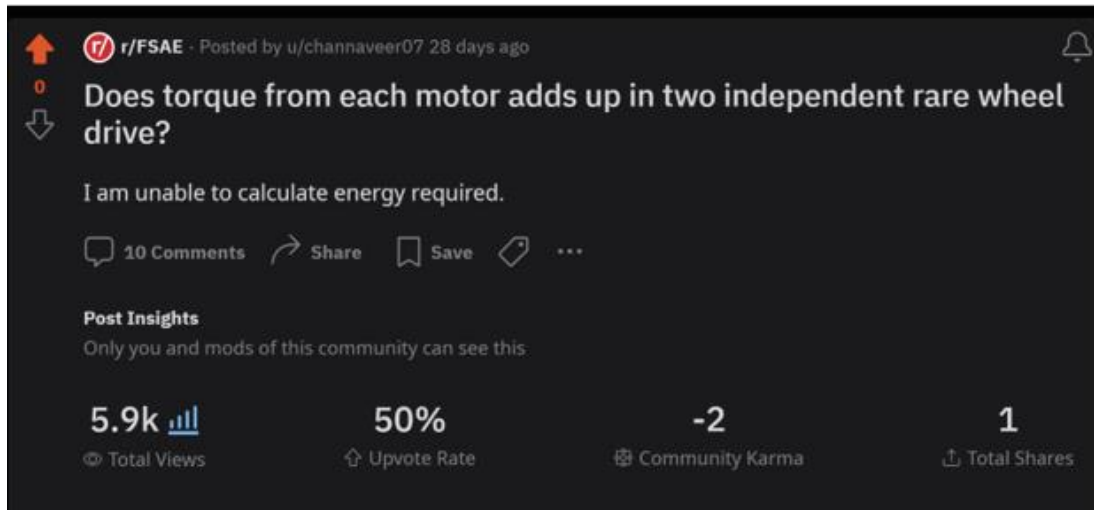


Fig.1.9 Survey graph 9

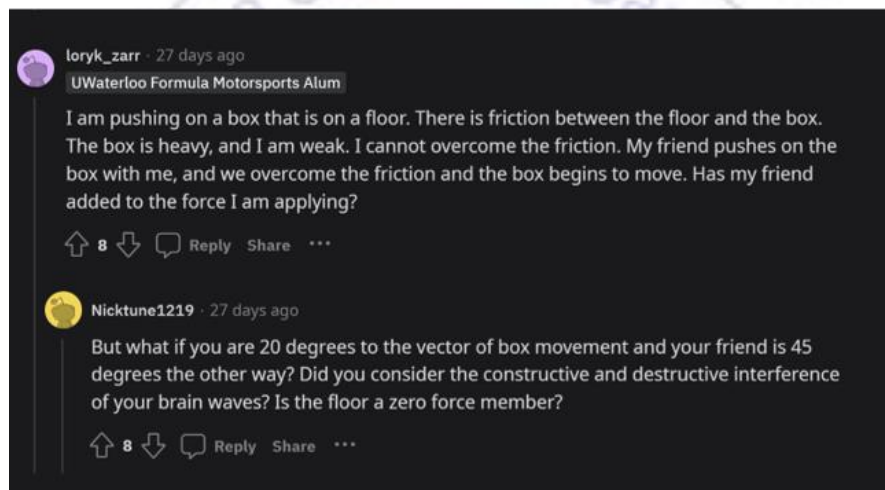


Fig.1.10 Survey graph 10

1.4 Survey Takeout:

- Many people haven't come across electronic differentials in vehicles before.
- **Perceived Benefits:**

Improved traction and stability: I believe improved traction and stability are crucial for safe and efficient driving, so I would rate this as a 4 out of 5.

Enhanced maneuverability: While important, I think maneuverability might not be as critical for all drivers, so I would rate it as a 3 out of 5.

Increased off-road performance: Given the potential benefits in off-road scenarios, I would rate this as a 4 out of 5.

- Better cornering capabilities: good cornering capabilities can significantly enhance driving experience, so I would rate this as a 4 out of 5.

Expectations:

Most people would expect an electronic differential system to effectively distribute power between the wheels, adapting seamlessly to various driving conditions. It should provide a noticeable improvement in stability and control, especially in challenging terrains.

Overall perception:

Most of the people quite likely to consider a vehicle equipped with an electronic differential system for my next purchase, given the potential benefits it can offer in terms of improved traction and stability.

1.5 Empathy Map:

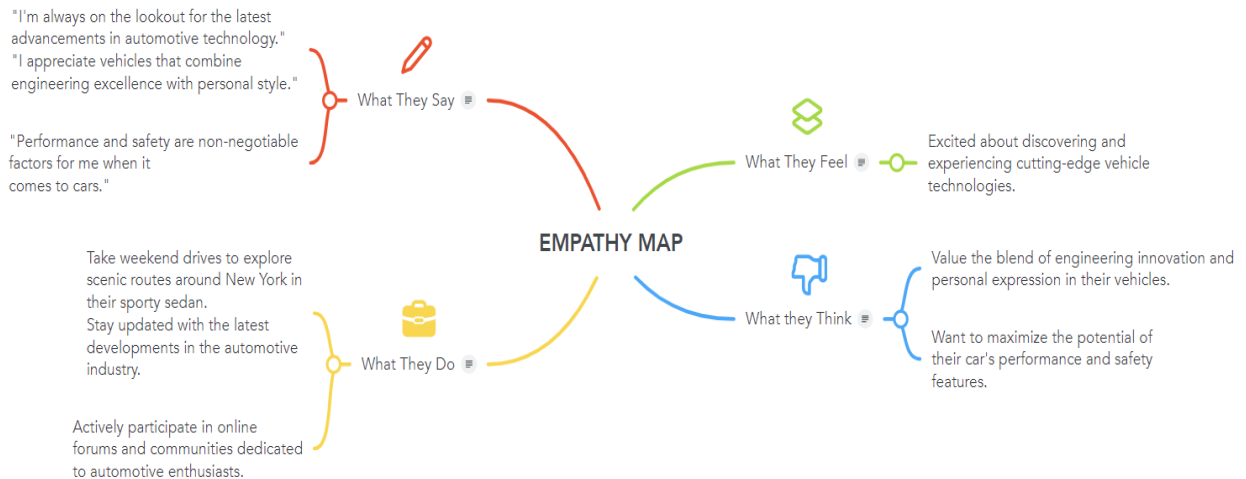


Fig 2.1. Empathy Map

1.6 Addressable Challenges and possible solutions:

Sensors and Data Integration:

Challenge: Ensuring accurate and real-time data collection from various sensors (wheel speed, vehicle speed, steering input, etc.).

Solution: Develop a robust sensor integration framework with appropriate filtering and calibration algorithms.

Control Algorithm Optimization:

Challenge: Designing a control algorithm that can efficiently distribute torque based on sensor inputs and driver commands.

Solution: Conduct extensive simulation and testing to fine-tune the control algorithm for optimal performance.

Hardware Reliability and Compatibility:

Challenge: Ensuring the electronic differential hardware is reliable, durable, and compatible with different vehicle architectures.

Solution: Rigorous testing under various conditions to identify and address potential hardware compatibility issues.

Dynamic Torque Management:

Challenge: Developing a system that can dynamically adjust torque distribution in real-time to adapt to changing driving conditions.

Solution: Implement advanced control strategies and algorithms to respond quickly to traction variations.

Scalability for Different Vehicle Types:

Challenge: Designing the prototype to be adaptable for integration into various types of vehicles, such as sedans, SUVs, or off-road vehicles.

Solution: Use modular design principles and ensure compatibility with different vehicle architectures.

Safety and Fail-Safe Mechanisms:

Challenge: Implementing fail-safe mechanisms to ensure that the system operates safely even in the event of sensor or hardware failures.

Solution: Integrate redundancy and monitoring systems to detect and mitigate potential failures.





CHAPTER 2

DEFINE

CHAPTER 2

DEFINE

2.1 Introduction to problem definition:

In the realm of vehicular dynamics, the conventional mechanical differentials have long been the cornerstone of torque distribution between the wheels. However, their limitations become evident in scenarios with varying traction conditions, leading to compromised stability, reduced manoeuvrability, and potential traction loss. To surmount these challenges and usher in a new era of precision in vehicle control, this project embarks on the development of a pioneering Electronic Differential System. This innovative system leverages advanced sensor technology and dynamic control algorithms to intelligently manage torque distribution in real-time, ensuring optimal traction and stability across a spectrum of driving environments.

The objective of this endeavour is to transcend the constraints of traditional differential systems and create a prototype that not only enhances vehicle performance but also presents a scalable solution adaptable to diverse vehicle platforms. Through meticulous design and integration, this Electronic Differential System promises to revolutionize the driving experience, particularly in scenarios demanding acute traction control, such as off-road expeditions, high-performance manoeuvres, and adverse weather conditions.

By delving into the intricacies of sensor integration, control algorithm optimization, and hardware compatibility, this project aspires to unlock a new dimension in vehicular engineering. The resulting prototype is envisioned to be a testament to the potential of electronic differentials in reshaping the landscape of automotive technology.

This introduction provides an overview of the project's goals, outlines the challenges it aims to address, and highlights the potential impact of the electronic differential prototype. It sets the tone for the reader to understand the significance and relevance of the project.

2.2 Ideation workflow

2.2.1 Task:

The conventional mechanical differentials in vehicles often face limitations in distributing torque effectively between the wheels, especially in scenarios with varying traction conditions.

This can lead to reduced maneuverability, traction loss, and compromised stability, particularly in off-road, slippery, or high-performance driving situations. To address these challenges, there is a pressing need for an advanced electronic differential system that can intelligently manage torque distribution between wheels in real-time

2.2.2 Objectives:

Safety Enhancement: Implement ACC to maintain safe following distances and LKA to ensure vehicles stay within their lanes, reducing the risk of accidents.

Improved Traffic Flow: Optimize vehicle speed based on traffic conditions, reducing congestion and improving overall traffic flow.

Driver Comfort and Convenience: Provide a comfortable driving experience by automating speed control and lane keeping, allowing drivers to focus on higher-level tasks.

Integration with Sensor Systems: Integrate ACC and LKA with advanced sensor systems for accurate perception of surrounding vehicles and road conditions.

2.2.3 Temporal Dependency:

The ACC system continuously monitors the relative speed and distance to the vehicle ahead in real-time.

The LKA system operates in real-time by tracking lane markings and making continuous steering adjustments.

2.2.4 Inter-Patient Variability:

Both systems rely on real-time sensor data to adapt to changing traffic conditions.

Variability in driving styles and preferences may require customizable settings for following distances and lane-keeping aggressiveness.

Different drivers may have varying comfort levels with the system's interventions, necessitating adjustable parameters.

2.2.5 Real-Time Processing:

Sensor data (e.g., radar, cameras) is processed in real-time to determine the distance to the leading vehicle and identify lane boundaries.

The control algorithms for ACC and LKA make instantaneous decisions based on this processed data.

The system continuously updates its control inputs to the vehicle's throttle, brake, and steering systems in real-time convenience, and efficiency in modern vehicles. It addresses challenges related to real-time data processing, adapting to driver preferences, and ensuring reliable performance in diverse traffic conditions.

2.3 DESIGN THINKING CHALLENGES IDENTIFIED

User-Centered Integration Challenge:

Challenge: Ensuring that the E Differential system integrates seamlessly into different vehicle models and architectures, accommodating various user preferences and driving styles.

Approach: Conduct user interviews, focus groups, and usability testing to understand user needs and preferences. Implement a modular design approach that allows for customization and adaptability.

Sensor Fusion and Data Processing Challenge:

Challenge: Effectively integrating data from multiple sensors (wheel speed, steering input, traction conditions) and processing it in real-time to make accurate torque distribution decisions.

Approach: Develop advanced sensor fusion algorithms that combine data from different sources for precise and reliable information. Implement robust real-time processing capabilities.

Dynamic Torque Management Challenge:

Challenge: Designing a control algorithm that dynamically adjusts torque distribution based on sensor inputs and driver commands, ensuring optimal traction and stability.

Approach: Utilize machine learning techniques and adaptive control strategies to create a responsive torque management system. Conduct extensive simulation and testing to fine-tune the algorithm.

Inter-Patient Variability Challenge:

Challenge: Accounting for variability in driver behaviour, preferences, and driving conditions, as different users may have varying levels of comfort and expectations from the E Differential.

Approach: Implement user-customizable settings for torque distribution parameters, allowing drivers to adjust the system to their individual preferences. Provide clear user interfaces for customization.

Safety and Fail-Safe Mechanisms Challenge:

Challenge: Ensuring that the E Differential system operates safely, even in the event of sensor or hardware failures.

Approach: Implement redundancy and monitoring systems to detect and mitigate potential failures. Develop fail-safe mechanisms that ensure safe operation in emergency situations.

Cost-Effectiveness and Affordability Challenge:

Challenge: Balancing the development costs with the affordability of the e Differential system for potential consumers.

Approach: Evaluate component costs, optimize manufacturing processes, and explore economies of scale for cost reduction. Consider scalability for mass production.

Regulatory Compliance and Certification Challenge:

Challenge: Ensuring that the E Differential system complies with relevant automotive safety and regulatory standards.

Approach: Collaborate with certification authorities and conduct thorough compliance testing throughout the development process. Stay informed about evolving industry standards and requirements.

These design thinking challenges highlight critical aspects to consider when developing an Electronic Differential (E Differential) system. Addressing these challenges through user-centered design, technical innovation, and rigorous testing will be key to creating a successful and reliable system.



CHAPTER 3

IDEATION

CHAPTER 3

IDEATION

The electronic differential represents a pioneering advancement in vehicle dynamics, revolutionizing traction and stability control. Departing from traditional mechanical differentials, it employs cutting-edge technology to intelligently distribute torque between wheels, optimizing handling on diverse road conditions. Particularly crucial for electric vehicles and those with independent rear-wheel drives, the electronic differential enhances stability during cornering and minimizes wheel slippage. This technology relies on sensors, actuators, and sophisticated control algorithms to ensure optimal torque distribution, enabling superior vehicle control and manoeuvrability. With its ability to adapt to varying driving scenarios, the electronic differential stands as a cornerstone of modern automotive engineering, providing safer and more dynamic journeys.

3.1 Introduction

In the realm of road transportation systems, the differential mechanism plays a pivotal role in preventing vehicles from skidding on curved pathways. In practical applications, conventional mechanical differentials have been utilized, but their sizable form contributes to increased weight and renders them less compatible with electric vehicles, especially those employing distinct propulsion systems for individual rear wheels. A recent and noteworthy advancement comes in the form of the electronic differential, which represents a technological leap in electric vehicle design, offering superior stability and control while navigating curved roads.

This study focuses on the modeling and simulation of an electronic differential that leverages an innovative wavelet transform (WT) controller for managing two brushless DC (BLDC) motors, each responsible for driving one of the rear wheels. In this novel approach, a discrete wavelet transform (DWT) controller is employed to dissect the discrepancy between the actual vehicle speed and the speed commanded by the electronic differential. This command is contingent on input parameters like throttle position and steering angle. The DWT controller dissects this speed error into distinct frequency components. Subsequently, by appropriately scaling these frequency constituents with their respective gains, a control signal is derived, which subsequently acts as the input to the motors.

The efficacy of this proposal was validated through a series of meticulously designed strategies, evaluated under different driving conditions: straight-line motion and both right and left turns. The simulation outcomes are presented and compared for both conventional PID controllers and the innovative WT controllers, assessing their robustness and stability in performance.

3.2 Graphical abstract

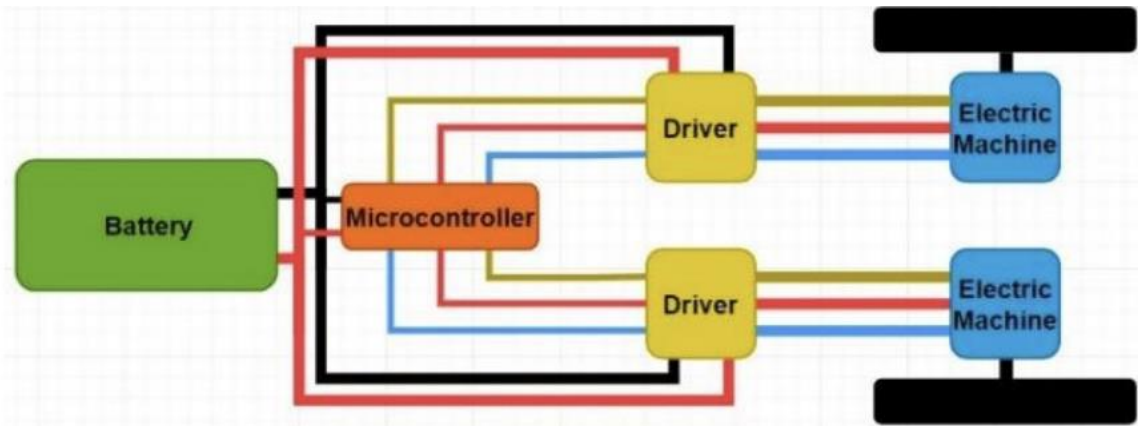


Figure 3.1 Fundamental structure of electric vehicle using electronic differential system

An electric vehicle (EV) features a simplified mechanical structure compared to internal combustion engine vehicles. It includes an electric motor that drives the wheels directly, eliminating the need for a complex transmission system. The absence of a conventional engine and exhaust components leads to a more spacious undercarriage. Additionally, EVs have a rechargeable battery pack that provides power to the motor, and regenerative braking systems help convert kinetic energy back into electrical energy, enhancing overall efficiency. This streamlined mechanical design contributes to reduced maintenance requirements and improved weight distribution, positively impacting EV performance.

3.3 Mechanical structure of an electric vehicle

An electric vehicle (EV) boasts a streamlined mechanical architecture in contrast to internal combustion engine counterparts. It integrates an electric motor that delivers direct propulsion to the wheels, obviating the necessity for intricate transmission assemblies. This departure from conventional powerplants translates to a more expansive undercarriage space. Moreover, EVs are equipped with a rechargeable battery pack, furnishing requisite power to the motor, while regenerative braking systems facilitate the conversion of kinetic energy into

electrical energy, engendering heightened overall energy efficiency. This optimized mechanical configuration not only curtails maintenance demands but also optimizes weight distribution, culminating in discernibly enhanced EV performance parameters.

3.3 System Modelling of Electronic Differential

The design of the electronic differential should enable the vehicle to navigate curves without experiencing any loss of traction. For this purpose, two inputs are the necessary functions, first steering angle and second throttle position. To avoid slipping and collapse of the vehicle, above stated two functions decides the speed of the right and the left wheel during propagation.

Fig. 1(a) illustrates the schematic of electronic differential proposed, where both the left and rights wheels are controlled by two separate motors. In case of right turn, the differential will have to retain left wheel at higher speed than the right wheel, hence prevent the tyres from losing traction on turning (right) and vice-versa (left) [18]. Fig. 1(b) illustrates the electric vehicle turn on curved roads .

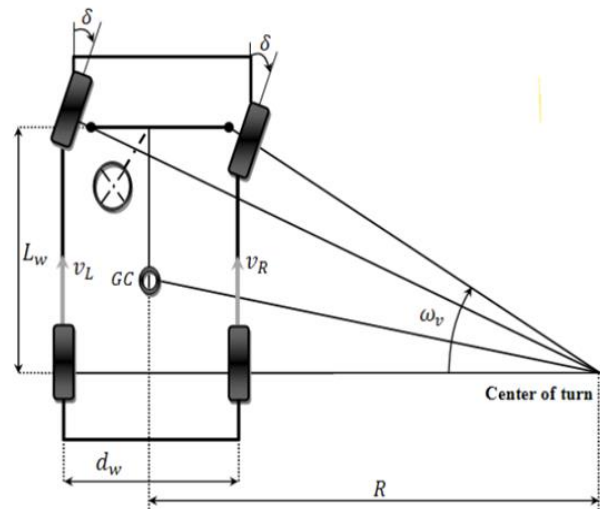


Figure 3.2 a) Schematic of electric vehicle with the electronic differential

Where,

L_w is the wheel base, δ is the turning angle, d_w is the track width, R is the radius of the turn and ω_R and ω_L represent the angular speeds of the left and the right wheel respectively. By the function of vehicle speed and the radius of the turn, the linear speed of each wheel is expressed as below:

$$v_R = \omega_v \left(R - \frac{d_w}{2} \right) \quad (2)$$

The expression relating the radius of the turn, steering angle and wheel base is given by:

$$R = \frac{L_w}{\tan \delta} \quad (3)$$

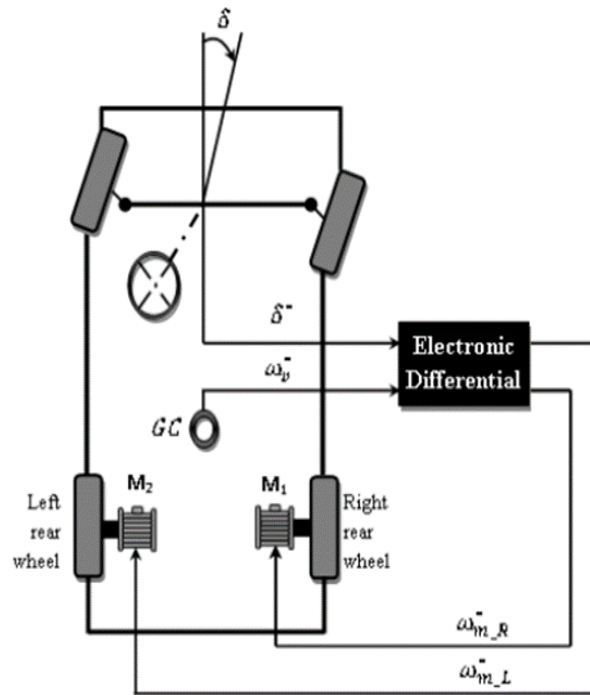


Figure 3.2 b) Schematic of the electric vehicle driven on curved road

By substitution Eq. 3 in Eq. 1 and Eq. 2, the angular speed of each wheel is express as:

$$\omega_{r_L} = \frac{L_w + \frac{1}{2}d_w \tan \delta}{L_w} \omega_v \quad (4)$$

$$\omega_{r_R} = \frac{L_w - \frac{1}{2}d_w \tan \delta}{L_w} \omega_v \quad (5)$$

Now, the difference in angular speeds between wheel drives is written as below:

$$\Delta\omega = \omega_{r_L} - \omega_{r_R} = \frac{d_w \tan \delta}{L_w} \omega_v \quad (6)$$

The direction of the turn is decided by the sign of the steering, which is actually given by:

- ☐ For turn right i.e. $\delta > 0$,
- ☐ For turn left i.e. $\delta < 0$,
- ☐ For straight ahead i.e. $\delta = 0$.

The driver provides the steering input, the electronic differential executes the immediate action by reduces the speed of the inner wheel and increases the speed of the outer wheel. Now, the driving speeds of the wheels are expressed as below:

$$\omega_{r_L}^* = \omega_v + \frac{\Delta\omega}{2} \quad (7)$$

$$\omega_{r_R}^* = \omega_v - \frac{\Delta\omega}{2} \quad (8)$$

BLDC motors are commonly employed in electric vehicles, with two prevalent configurations: interior rotor and outer rotor types, both featuring a rotor comprised of permanent magnets. The number of magnet pieces corresponds to the motor's pole count, while phase windings are strategically positioned on the stator to generate a trapezoidal back-EMF shape. This design yields advantages such as enhanced efficiency, greater torque density, quiet operation, and reduced maintenance requirements. The BLDC motors play a pivotal role in propelling electric vehicles. The mathematical model of a three-phase BLDC motor, as illustrated in Figure 2, is encapsulated by the following equation:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L-M & 0 & 0 \\ 0 & L-M & 0 \\ 0 & 0 & L-M \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix}$$

Where,

$V_{a,b,c}$ are the three-phase voltage,

$i_{a,b,c}$ are the three-phase current,

$e_{a,b,c}$ are the phase back-EMF,

R, L are the stator resistance and stator inductance,

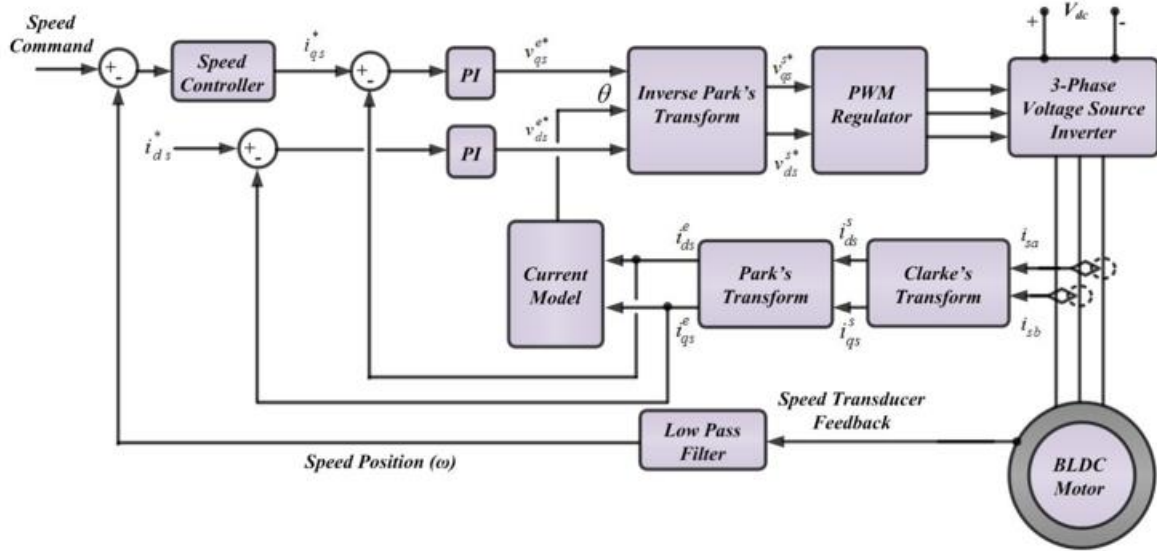


Fig. 3.3. Schematics of indirect field-oriented control (IFOC) of BLDC motor drive.

3.4 Wavelet Transform based Speed Controller

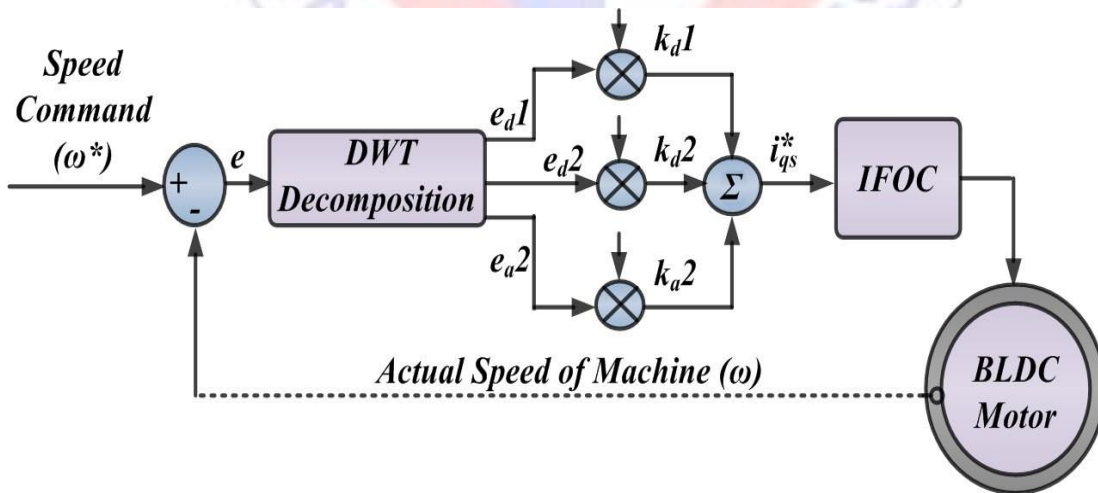


Fig. 3.4. Schematics of IFOC wavelet speed control scheme for BLDC motor.

Figure 3.4 provides a comprehensive schematic overview of the WT speed controller for a BLDC motor drive [11-14]. To ensure precise speed regulation for both the right and left motors, a Wavelet Transform (WT) is applied to each speed controller within the two BLDC motor drives. The choice of WT type and the number of decomposition levels serve as two key criteria for the WT controller.

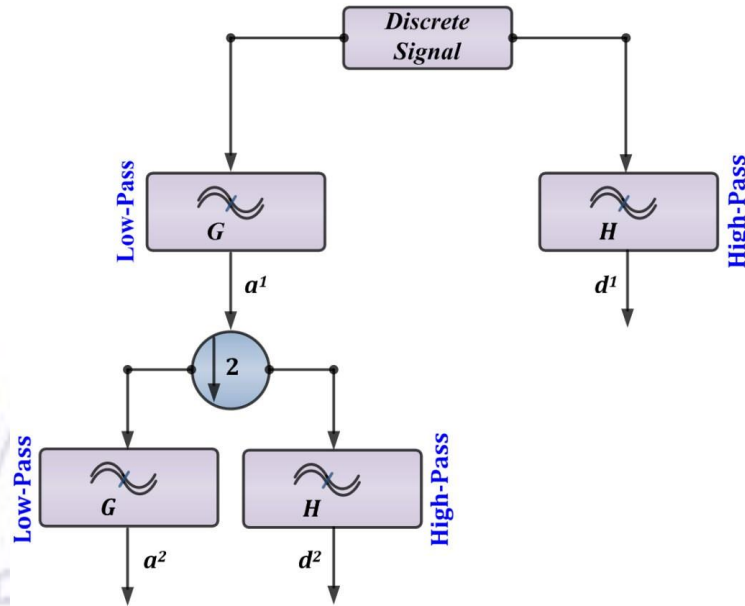


Fig. 3.5. Schematics of DWT 2-level decomposition algorithm.

The Discrete Wavelet Transform (DWT) offers a time-scale representation of digital signals through the application of digital filtering techniques. The resolution of the DWT is modified through filtering operations, while the scales are adjusted through down-sampling operations. A typical example of a 2-level decomposition tree is depicted in Figure 4. The decomposition process initiates when a discrete signal $x[n]$ of length N undergoes high-pass filtering, yielding an impulse response $h[n]$, and low-pass filtering, resulting in an impulse response $g[n]$. A one-level decomposition of the DWT is constituted by the outputs of these high and low pass filters, and it can be mathematically expressed as follows [10-14, 19-20]:

$$d^1[n] = \sum_{k=0}^{N-1} x[k]h[n-k] \quad (9)$$

$$a^1[n] = \sum_{k=0}^{N-1} x[k]g[n-k]. \quad (10)$$

Where $d1[n]$ and $a1[n]$ represent the outputs of the high-pass and low-pass filters respectively. Following this, the output from the low-pass filter undergoes down-sampling by a factor of two and is subsequently processed through both a low-pass and a high-pass filter, akin to those utilized in the 1-level decomposition. The mathematical expression for the 2-level decomposition of the discrete signal after down-sampling is as follows:

$$d^2[n] = \sum_{k=0}^{N/2-1} a^1[k] h[n-k] \quad (11)$$

$$a^2[n] = \sum_{k=0}^{N/2-1} a^1[k] g[n-k] \quad (12)$$

Selection of Wavelet function

There exist various types of Wavelet Transforms (WT), and the choice depends on how well the signal aligns with the specific application. In this proposed investigation, the WT analysis is intended to be employed on the error signal of the BLDC drive. To this end, the minimum description length (MDL) data criterion has been chosen as the most suitable WT filter. The optimal quantity of WT coefficients to retain for signal reconstruction can be mathematically defined as follows [19]:

$$MDL(k, n) = \min \left\{ \frac{3}{2} k \log N + \frac{N}{2} \log \left\| \tilde{\alpha}_n - \alpha_n \right\|^2 \right\}, \quad (13)$$

$$0 < k < N; \quad 1 \leq n \leq M$$

where, $\tilde{\alpha}_n = Wnf$ denotes a vector of the WT transformed coefficients of the signal f using WT

In the given context, the notation $\alpha(k) = \mathcal{O}K\tilde{\alpha}_n = \mathcal{O}K(Wnf)$ signifies a vector containing 'k' nonzero elements. The threshold parameter $\mathcal{O}K$ retains 'k' of the largest elements in the vector $\tilde{\alpha}_n$ while setting all other elements to zero. Here, 'N' and 'M' represent the length of the signal and the number of Wavelet Transform (WT) filters, respectively.

In this investigation study, the orthogonal WT filter 'db4' of Daubechies has been chosen for implementation in the IFOC speed controller based on the Minimum Description Length (MDL) criterion.

Levels of Decomposition

The number of levels of decomposition represents the gains of the controller and for deducing this; an entropy-based criterion was used. The entropy $H(x)$ of a signal $x[n]$ of length N is defined as below [10, 19]:

$$H(x) = - \sum_{n=0}^{N-1} |x(n)|^2 \log |x(n)|^2. \quad (14)$$

To determine the optimal levels of decomposition, the entropy is evaluated at each level and a new level j , if as follows:

$$H(x)_j \geq H(x)_{j-1} \quad (15)$$

Further, the level j is omitted from the DWT tree. For this proposed investigation, 2-levels of decomposition are more sufficient and effective to represent error signal [11-14].

Wavelet Controller for Electronic Differential of Electric Vehicle

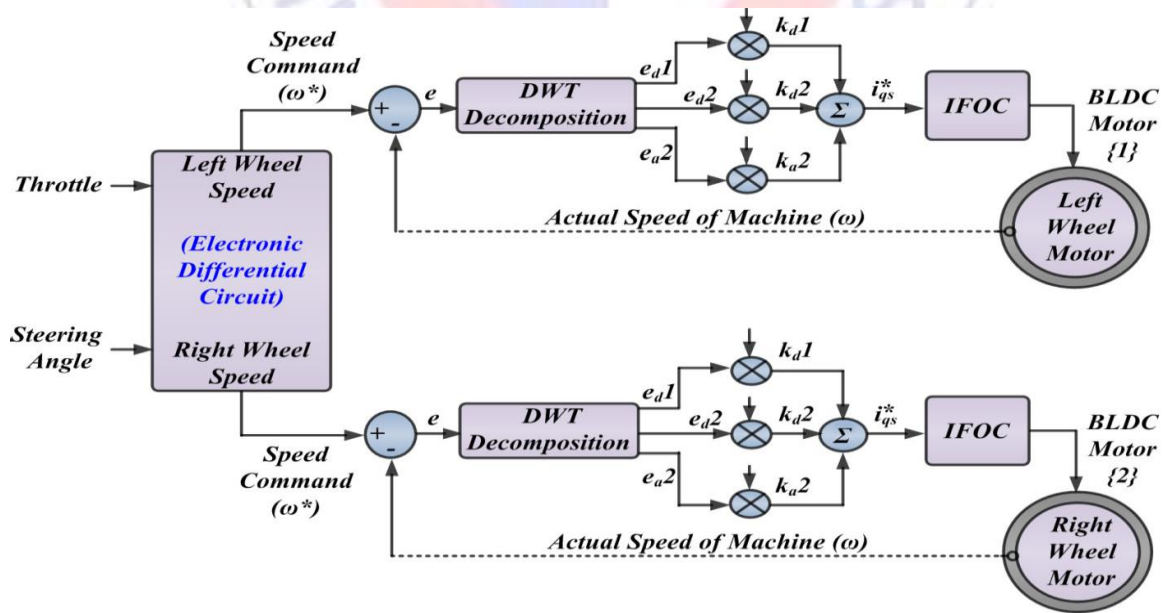


Fig. 3.6. Schematics of the proposed novel WT based IFOC for electronic differential of two BLDC motors.

Figure 3.6 illustrates the control schematic of the WT IFOC (Indirect Field-Oriented Control) speed controller for two BLDC motors connected to the rear wheels of the electric vehicle. The input parameters provided are the throttle position and steering angle, which are fed into the electronic differential. This differential system calculates the desired speeds for both the left and right motors. The error detector then compares these desired speeds with the actual speeds, generating an error speed signal. This error speed signal is subsequently utilized by the WT controllers to produce the control signals for the IFOC drive systems of the two BLDC motors.

Through the WT Controller, the error signal undergoes decomposition into its detailed (high-frequency) and approximated (low-frequency) components [11-14, 17, 21-22]. These components are then scaled by their respective gains and combined to generate the control signal 'u', according to the following equation

$$u = k_{d1}e_{d1} + k_{d2}e_{d2} + \dots + k_{dN}e_{dN} + k_{aN}e_{aN} \quad (16)$$

where, gains k_{d1} , k_{d2} , ..., k_{dN} are used to tune the high and medium frequency components of the error signal (e_{d1} , e_{d2} , ..., e_{dN}). Whereas, gain k_{aN} is used to tune the low frequency components of the error signal (e_{aN}) and N is the number of decomposition levels.

To be noted that the gain values of approximation coefficients (k_{aN}) are responsible for controller functioning. Therefore, lesser approximation gain value lesser the peak overshoot. The gain values of detailed coefficients (k_{d1} , k_{d2} ... k_{dN}) are only used for controlling high frequency signals which are produced in the system due to sensor noise signals and these gain values do not affect the output speed in ideal noise free conditions [13-14, 21-22].



CHAPTER 4

PROTOTYPE

CHAPTER 4

PROTOTYPE

4.1 Prototype Development Process:

The prototyping phase of the differential system was a meticulously planned endeavour, involving the creation of a scaled-down, functional model that emulates the core functionality of the final product. This process enabled engineers to iteratively refine and test various components, ensuring seamless integration into different vehicle architectures.

4.1.1 Component Testing and Validation:

Each component of the differential prototype underwent rigorous testing to validate its performance and functionality. This included the evaluation of sensor accuracy, control algorithm responsiveness, and real-time torque distribution capabilities. Through this iterative testing process, potential improvements and optimizations were identified and implemented.

4.1.2 Sensor Integration and Calibration:

The prototype's success hinged on the precise integration and calibration of various sensors, including wheel speed sensors, steering angle sensors, and traction sensors. Calibration procedures were developed to ensure accurate data acquisition, allowing the system to make informed decisions in real-time.

4. Control Algorithm Fine-Tuning:

The heart of the E-Differential system lay in its control algorithm, which determined torque distribution based on sensor inputs and driver commands. Extensive simulation and testing were conducted to fine-tune the algorithm, optimizing its responsiveness to varying driving conditions.

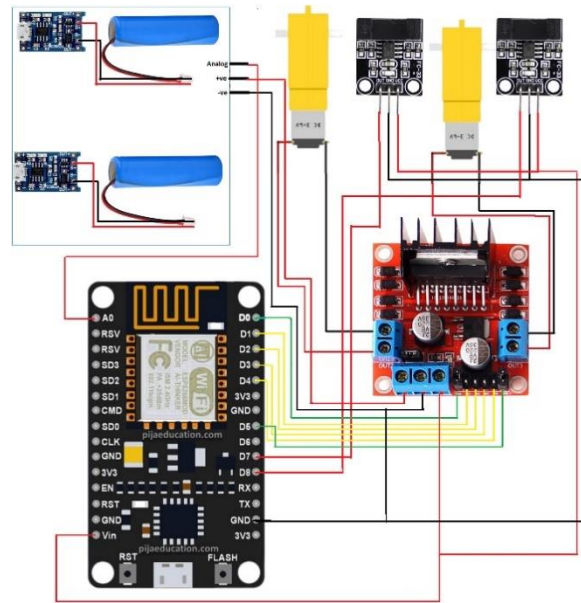


Fig.4.1 Circuit diagram of the control system

Components used and technologies used:

- **Encoders:** Encoders used in wheels, often referred to as "wheel encoders" or "rotary wheel encoders," are devices that measure the rotational motion and position of a wheel
- **Motor Driver:** The L298 is a dual full-bridge motor driver integrated circuit (IC) commonly used to control the speed and direction of DC motors and stepper motors in robotics and automation applications.
- **Charging Module:** The TP4056 charging module is a compact device that charges lithium-ion or lithium-polymer batteries efficiently and safely, with features like constant current and voltage charging, protection against overcharging, and LED indicators for charging status.
- **Battery:** The 18650 lithium-ion battery is a cylindrical cell format often used in various electronic devices, including laptops and power banks. Its specifications typically include a nominal voltage of 3.7V, capacity ranging from around 1500mAh to over 3500mAh, and a rechargeable design
- **Microcontroller:** The ESP8266 is a Wi-Fi-enabled microcontroller module used for IoT and embedded applications. It enables wireless communication, sensor integration, web server hosting, and IoT device control.

- **Motors:** A DC geared motor is a combination of a DC motor and a gearbox, providing controlled torque and speed output. It's used in various applications like robotics and automation to convert electrical energy into mechanical motion with adjustable speed and power.
- **Controlling:** Arduino Cloud is a user-friendly platform enabling remote control and monitoring of Arduino-based projects from any location. It offers secure connectivity, data visualization, automation capabilities, and supports multiple devices, making it ideal for both hobbyists and IoT users.

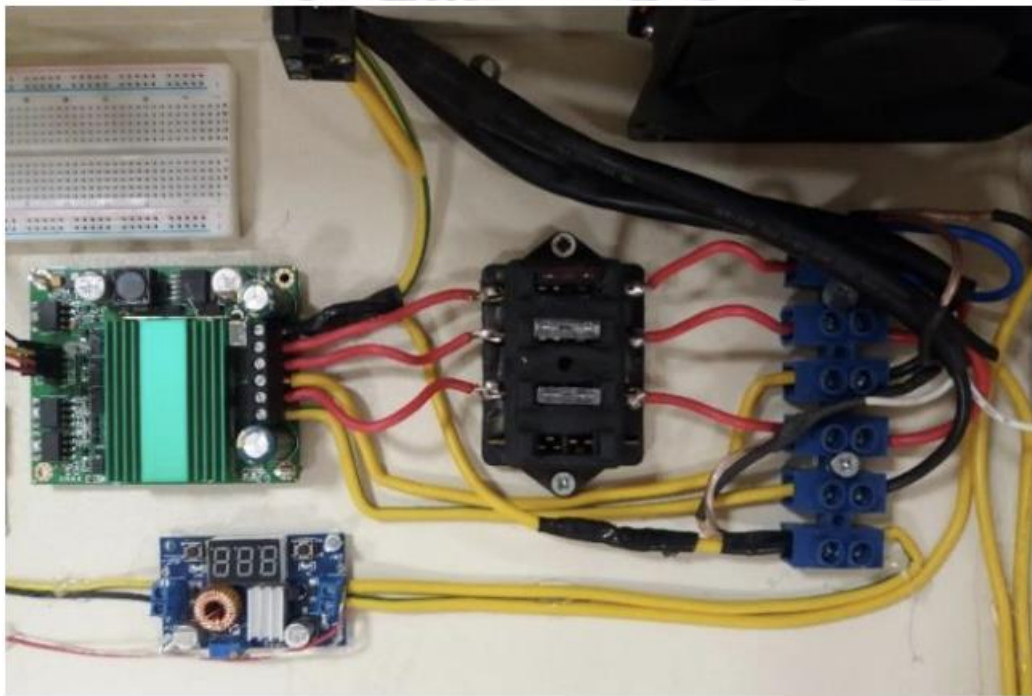


Fig 4.2 Final Prototype



CHAPTER 5 TESTING

CHAPTER 5

TEST RESULT ANALYSIS

5.1 RESULTS OF THE SIMULATION

The speed data of the wheels obtained as a result of the road tests carried out are:

	Left wheel	Right wheel
5°	%1.46	%1.46
15°	%1.70	%1.70
25°	%3.40	%3.40
Average	%2.19	%2.19
Difference	%0.00	

Table5.1: Slip values according to the steering angle for slippery test area

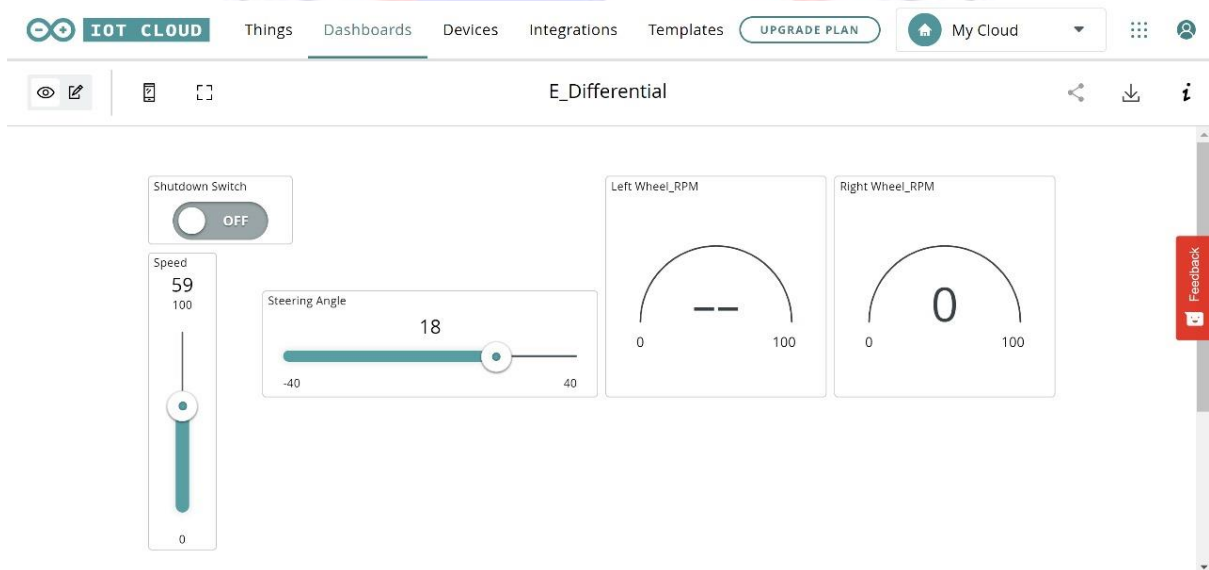


Fig.5.2 Watch window of Arduino IoT cloud

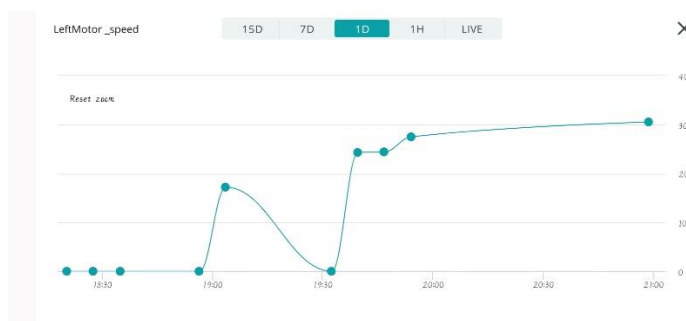


Fig.5.3 Output of left motor speed in Arduino output window

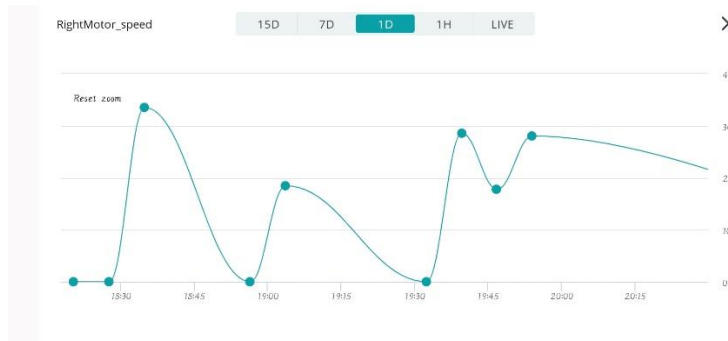


Fig.5.4 Output of right motor speed in Arduino output window

5.2 ANALYSIS:

When the graphics were examined independently from the steering angle, the EDS control was ensured the control of system in all road conditions. As well as, it is experimentally observed that as the steering angle increases, the difference in speed between the right and left wheels increases.

$$s = |v_{AJT} - v_{real}| / v_{AJT} \quad (7)$$

In eq. (7), the mathematical expression of the slip in wheel speed for vehicles is given. This expression is one of the most important parameters that express whether the speed of vehicle catches to the reference speed or not. In line with the tests performed with the EDS control, the average slip value was calculated for three different test areas and three different steering angles by using eq. (7). Afterwards, the average slip value for each test area was obtained by averaging the data on the same test area but at different steering angles. As a result, the slip values according to the state of test area and steering angles are given in tables.

When the slip data given in Tables are examined, it is seen that the EDS control used is effective in all road conditions considered in experimental studies. Therefore, the designed electric vehicle was able to turn the bends stably. The current data of DC motors driven the wheels were continuously measured and recorded during experimental studies. Electric motor current data are given in figures 9, 10 and 11



CHAPTER 6

CONCLUSION AND FUTURE SCOPE

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The development of the Electronic Differential (e-Differential) system marks a significant milestone in automotive engineering, poised to redefine the way vehicles navigate a diverse range of driving scenarios. Through an intricate fusion of cutting-edge sensor technology and dynamic control algorithms, the e-Differential prototype has showcased its remarkable ability to intelligently distribute torque in real-time, ensuring optimal traction and stability. This breakthrough promises to revolutionize vehicle dynamics, transcending the limitations of conventional mechanical differentials. From navigating challenging off-road terrains to delivering precise control in high-speed cornering, the e-Differential system demonstrates its adaptability and versatility.

One of the standout features of the e-Differential lies in its user-centric design, emphasizing personalization and comfort. By allowing drivers to customize torque distribution preferences, the system caters to the diverse array of driving styles and preferences. This level of adaptability ensures that the e-Differential seamlessly integrates into a wide spectrum of vehicles and becomes an intuitive extension of the driver's intent. Furthermore, the integration of fail-safe mechanisms and redundant sensor systems provides an added layer of assurance, instilling confidence in the system's reliability, even in unpredictable and demanding driving conditions.

As we gaze towards the horizon of autonomous driving and increasingly complex traffic environments, the E-Differential system emerges as a pioneering force. It not only meets the technical challenges of electronic differential development but also underscores a fundamental shift towards human-centric engineering. The successful integration of this groundbreaking technology is a testament to the collaborative efforts of engineers, designers, and researchers dedicated to pushing the boundaries of automotive technology.

In conclusion, the E-Differential system represents a transformative leap in automotive engineering, promising safer, more efficient, and more enjoyable driving experiences. With its potential to enhance traction, stability, and maneuverability, it is poised to become an integral component of the vehicles of the future. The success of this project serves as an inspiration for

future endeavors, reaffirming the boundless possibilities that await in the realm of vehicle dynamics and electronic drivetrain technology.

6.2 FUTURE SCOPE

The successful development and integration of the Electronic Differential (E-Differential) system open up a wide array of future possibilities and avenues for further exploration. As automotive technology continues its rapid evolution, the E-Differential system is poised to play a pivotal role in shaping the future of vehicular dynamics.

One of the most exciting prospects lies in the seamless integration of the E-Differential system into autonomous driving platforms. With the ability to adapt torque distribution in real-time, the E-Differential can serve as a vital component of advanced driver-assistance systems (ADAS). This integration holds the potential to significantly enhance the vehicle's ability to navigate complex and dynamic traffic environments, further paving the way for the widespread adoption of autonomous driving.

The incorporation of machine learning algorithms represents another exciting frontier for the E-Differential system. By leveraging artificial intelligence, the system can learn and adapt to specific driver preferences and driving conditions over time. This dynamic adaptation can lead to even more personalized and intuitive driving experiences, ensuring that the vehicle's performance aligns seamlessly with the driver's individual style and preferences.

Furthermore, the E-Differential system stands to make a substantial impact on the energy efficiency of vehicles, particularly in the realm of hybrid and electric vehicles (EVs). By optimizing torque distribution, the system can contribute to improved energy efficiency and extended range. This has the potential to revolutionize regenerative braking systems and overall drivetrain efficiency, further advancing the sustainability and environmental impact of automotive technology.

In anticipation of future advancements in sensor technologies, the E-Differential system can benefit from the integration of more advanced sensors, such as LiDAR and state-of-the-art camera systems. These sensors provide even more precise and detailed information for the E-Differential system to process, allowing for finer-tuned adjustments in torque distribution. This enhanced sensor suite will play a pivotal role in further elevating the system's adaptability and responsiveness to a diverse range of driving conditions.

As the automotive industry embraces increased connectivity, ensuring the security of the E-Differential system becomes paramount. Implementing robust cybersecurity measures and enabling over-the-air updates will be crucial in safeguarding the system against potential vulnerabilities. This proactive approach to cybersecurity will be essential in maintaining the integrity and reliability of the E-Differential system in an increasingly connected automotive landscape.

Collaboration with other automotive manufacturers and industry stakeholders is another avenue of future development for the E-Differential system. By working together, the industry can establish standards for electronic drivetrain technologies, ensuring compatibility and interoperability across different vehicle models and brands. This collaborative effort will not only accelerate the adoption of electronic drivetrain innovations but also drive forward the collective progress of the automotive industry as a whole.

In conclusion, the future scope for the E-Differential system is not only technologically promising but also encompasses a broader vision for safer, more efficient, and more enjoyable driving experiences. As research and development continue to push boundaries, the potential applications and benefits of this system are poised to grow exponentially. The E-Differential system stands as a testament to the relentless pursuit of innovation in the automotive industry, and its future impact promises to be nothing short of transformative.

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APPENDIX PROGRAMS EXECUTED

APPENDIX

PROGRAMS EXECUTED

Arduino IDE Code:

```
const int leftEncoderPin = 15;

const int rightEncoderPin = 13;

volatile unsigned int leftEncoderCount = 0;

volatile unsigned int rightEncoderCount = 0;

unsigned long lastTime = 0;

/*

Sketch generated by the Arduino IoT Cloud Thing "Untitled"

https://create.arduino.cc/cloud/things/b6e22bd4-c9f7-4340-bc88-96bdc7e7abe3

Arduino IoT Cloud Variables description

int sa;

int smtl;

int smtr;

int speed;

bool sw;

These functions are generated with the Thing and added at the end of this sketch.

#include "thingProperties.h"

void countLeftEncoderPulse() {

    leftEncoderCount++;

}

void countRightEncoderPulse() {

    rightEncoderCount++;
```

```
}
```

```
void setup() {  
  
    pinMode(leftEncoderPin, INPUT_PULLUP);  
  
    pinMode(rightEncoderPin, INPUT_PULLUP);  
  
    attachInterrupt(digitalPinToInterrupt(leftEncoderPin), countLeftEncoderPulse, FALLING);  
  
    attachInterrupt(digitalPinToInterrupt(rightEncoderPin), countRightEncoderPulse,  
FALLING);  
  
    // Initialize serial and wait for port to open:  
  
    Serial.begin(9600);  
  
    pinMode(16, OUTPUT); //ena  
  
    pinMode(5, OUTPUT);  
  
    pinMode(4, OUTPUT);  
  
    pinMode(0, OUTPUT);  
  
    pinMode(2, OUTPUT);  
  
    pinMode(14, OUTPUT); //ena  
  
    // This delay gives the chance to wait for a Serial Monitor without blocking if none is found  
    delay(1500);  
  
  
    // Defined in thingProperties.h  
  
    initProperties();  
  
  
    // Connect to Arduino IoT Cloud  
  
    ArduinoCloud.begin(ArduinoIoTPreferredConnection);  
  
    /*
```

The following function allows you to obtain more information related to the state of network and IoT Cloud connection and errors the higher number the more granular information you'll get.

The default is 0 (only errors).

Maximum is 4

*/

```
setDebugMessageLevel(2);
```

```
ArduinoCloud.printDebugInfo();
```

```
}
```

```
void loop() {
```

```
  ArduinoCloud.update();
```

```
  unsigned long currentTime = millis();
```

```
  if (currentTime - lastTime >= 1000) {
```

```
    noInterrupts();
```

```
    smtl = (2 * PI * leftEncoderCount) / (20);
```

```
    smtr = (2 * PI * rightEncoderCount) / (20);
```

```
    leftEncoderCount = 0;
```

```
    rightEncoderCount = 0;
```

```
    interrupts();
```

```
    lastTime = currentTime;
```

```
  }
```

```
  int m;
```

```
  int n;
```

```
if(sw==true){  
  
int l=abs(speed*(1-tan(sa*3.14/180)));  
  
int r=abs(speed*(1+tan(sa*3.14/180)));  
  
if(l==0){m=0;}  
  
else if(l>=1 && l<10){m=500;}  
  
else if(l>=1 && l<20){m=530;}  
  
else if(l>=20 && l<30){m=560;}  
  
else if(l>=30 && l<40){m=590;}  
else if(l>=40 && l<50){m=620;}  
else if(l>=50 && l<60){m=650;}  
else if(l>=60 && l<70){m=680;}  
else if(l>=70 && l<80){m=710;}  
else if(l>=80 && l<90){m=740;}  
else if(l>=90 && l<100){m=770;}  
else if(l>=100 && l<110){m=800;}  
else if(l>=110 && l<120){m=830;}  
else if(l>=120 && l<130){m=860;}  
else if(l>=130 && l<140){m=890;}  
  
else if(l>=140 && l<150){m=920;}  
  
else if(l>=150 && l<160){m=950;}  
  
else {m=1024;}  
  
  
if(r==0){n=0;}  
  
else if(r>=1 && r<10){n=500;}
```

```
else if(r>=10 && r<20){n=530;}
else if(r>=20 && r<30){n=560;}
else if(r>=30 && r<40){n=590;}
else if(r>=40 && r<50){n=620;}
else if(r>=50 && r<60){n=650;}
else if(r>=60 && r<70){n=680;}
else if(r>=70 && r<80){n=710;}
else if(r>=80 && r<90){n=740;}
else if(r>=90 && r<100){n=770;}
else if(r>=100 && r<110){n=800;}
else if(r>=110 && r<120){n=830;}
else if(r>=120 && r<130){n=860;}
else if(r>=130 && r<140){n=890;}
else if(r>=140 && r<150){n=920;}
else if(r>=150 && r<160){n=950;}
else {n=1024;
digitalWrite(5,HIGH);
digitalWrite(4,LOW);
analogWrite(16,m);
digitalWrite(0,HIGH);
digitalWrite(2,LOW);
analogWrite(14,n);
}
```

```
else{  
    digitalWrite(4,LOW);  
    digitalWrite(5,LOW);  
    analogWrite(16,LOW);  
    digitalWrite(0,LOW);  
    digitalWrite(2,LOW);  
    analogWrite(14,LOW);  
}  
}  
/*  
    Since Speed is READ_WRITE variable, onSpeedChange() is  
    executed every time a new value is received from IoT Cloud.  
*/  
void onSpeedChange() {  
    // Add your code here to act upon Speed change  
}  
/*  
    Since Sw is READ_WRITE variable, onSwChange() is  
    executed every time a new value is received from IoT Cloud.  
*/  
void onSwChange() {  
    // Add your code here to act upon Sw change  
}
```

/*

Since Sm is READ_WRITE variable, onSmChange() is
executed every time a new value is received from IoT Cloud.

*/

```
void onSmChange() {
```

```
    // Add your code here to act upon Sm change
```

```
}
```

/*

Since Sa is READ_WRITE variable, onSaChange() is
executed every time a new value is received from IoT Cloud.

*/

```
void onSaChange() {
```

```
    // Add your code here to act upon Sa change
```

```
}
```

/*

Since Smtr is READ_WRITE variable, onSmtrChange() is
executed every time a new value is received from IoT Cloud.

*/

```
void onSmtrChange() {
```

```
    // Add your code here to act upon Smtr change
```

```
}
```

/*

Since Smtl is READ_WRITE variable, onSmtlChange() is
executed every time a new value is received from IoT Cloud.

*/

```
void onSmtlChange() {
```

```
    // Add your code here to act upon Smtl change
```

```
}.
```

